Flexible GLMM Toolbox

Upload Data Frame (CSV or Excel)

Browse... glmm_tutorial_dataset.csv

Upload complete

Select columns to include or exclude from analysis:

Select Columns to Keep:

- [x] Subject
- [x] Reaction_time
- [x] Days
- [x] age
- [x] bmi
- [x] Sex
- [x] Chronotype
- [x] Total_sleep_time
- [ ] Caffeine

Apply Column Selection

Categorical Variables

Numeric Variables

Apply Variable Types

Missing Value Treatment

(•) Remove rows with NA

Remove Missing Values

```
Before: 400 x 8
```

```
After: 400 x 8
```

```
Missing per variable: Subject: 0, Reaction_time: 0, Days: 0, age: 0, bmi: 0, Sex: 0, Chronotype: 0, Total_sleep_time: 0
```

Outlier Detection Method

None

SD cutoff (Z-score)

Cook's Distance cutoff (default = 4/n)

Mahalanobis cutoff (Chi-square quantile, e.g. 0.99)

0,99

Remove Outliers

Download Cleaned CSV (session/8aa814ed4947d58702a219a28d8a1317/download/download_no_outliers?w=eee636347ef44974bd62634a7aa0a479)

Select Variables to Standardize

Numeric preprocessing

- ( ) Center and scale
- (•) Center only
- ( ) Scale only
- ( ) None

Apply data standardization | View Processed Data

Multiple y and x inputs are allowed

Dependent Variable (y)

Reaction_time

For fit distribution check dependent variable must be a numeric

Fit Distribution

```
Distribution fitting complete.
```

Independent Variables (x)

Days

Interaction Variables

Interaction Mode

- (•) None
- ( ) IV * Interactor1
- ( ) IV * Interactor1 * Interactor2 (3-way)

Covariates (main effects)

age bmi Sex Total_sleep_time

GLMM Family

gaussian

Link Function

default

Modeling Engine

afex::mixed

Correlation structure (nlme)

None

Random Effects (lme4 / afex syntax)

(1|Subject)

Random Effects (nlme syntax)

~1|Subject

Optional Random Grouping variable for Auto correlation (e.g., Subject)

Optional Time variable for AR structures (e.g., trial, block, day, session)

Factors for Post-hoc (EMMs)

Sex

Custom model equation (overrides auto)

Run Models

Instructions | Data | FitDist Output | Fit Distribution Plots | Model Output | ANOVA | Summary Table | Performance | Post-hoc (EMMs) | Summary Plots | Diagnostics | Auto Report

The current version of Flexible GLMM toolbox can be used to

- fit GLMM models using afex and glmer R packages
- get results similar to SAS outputs
- identify fit distribution family for the dependent variables
- run analysis for a multiple dependent and independent variables simultaneously

Usage:

- Data input format: xlsx or csv file with header row containing variable names.
- By default first sheet will be used as the input. Ex. mtcars, sleepstudy (lme4)
- Missing values are blank/empty cells in the data
- For outlier removal specify the standard deviation value (ex. 3)
- Multiple covariates can be selected from the input data (age sex bmi)
- For family of distributions Refer- https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/family
- Custom equation format : y ~ x1 + x2

Note:

- Afex mixed - Refer to https://cran.r-project.org/web/packages/afex/afex.pdf
- For using complex models use custom equation option.
- For feedback/queries, please send an email to ptalwar@uliege.be; talwar.puneet@gmail.com.

Random effect structure in GLMM

Formula (lme4)	Alternative (lme4)	Formula (nlme)	Meaning
(1 | g)	-	~ 1 | g	Random intercept with fixed mean
(1 | g1:g2)	-	~ 1 | g1:g2	Random intercept for each level of the interaction between g1 and g2
(1 | g1/g2)	(1 | g1) + (1 | g1:g2)	~ 1 | g1/g2	Intercept varying among g1 and g2 within g1 (g2 Nested within g1)
(1 | g1) + (1 | g2)	-	-	Intercept varying among g1 and g2 (Crossed random intercepts)
(x | g)	(1 + x | g)	~ x | g	Correlated random intercept and slope
(x || g)	(1 | g) + (0 + x | g)	list(g = pdDiag(~ x))	Uncorrelated random intercept and slope
(0 + x | g)	-	~ x - 1 | g	Uncorrelated random slope/Random slope for **x** only (no intercept)

g, g1, g2 are **categorical grouping variables**, while **x** is typically a **continuous predictor**.

Data View

Show 10 entries Search:

	Subject	Reaction_time	Days	age	bmi	Sex	Chronotype		Tota
1	S01	344.7975609	0	65	22.9	Male	Evening	6.558002389	1
2	S01	358.2948244	1	65	22.9	Male	Evening	5.694196873	1
3	S01	346.7573789	2	65	22.9	Male	Evening	6.080514593	0
4	S01	348.7688488	3	65	22.9	Male	Evening	8.695730718	2
5	S01	405.1083666	4	65	22.9	Male	Evening	5.788823549	4
6	S01	410.7200691	5	65	22.9	Male	Evening	6.691790804	0
7	S01	474.0476396	6	65	22.9	Male	Evening	7.118221633	2
8	S01	436.0239845	7	65	22.9	Male	Evening	6.970580693	1
9	S01	471.6682438	8	65	22.9	Male	Evening	8.571669568	2
10	S01	479.0655283	9	65	22.9	Male	Evening	6.972329237	3
11	S02	253.3418117	0	31	28.6	Male	Evening	5.513699118	1
12	S02	280.6441681	1	31	28.6	Male	Evening	6.76515959	2
13	S02	297.8832861	2	31	28.6	Male	Evening	6.9153421	3
14	S02	303.0826851	3	31	28.6	Male	Evening	5.669800307	2
15	S02	287.7496234	4	31	28.6	Male	Evening	7.994339902	2
16	S02	315.0243614	5	31	28.6	Male	Evening	7.31314609	2
17	S02	375.3720666	6	31	28.6	Male	Evening	8.877980186	3
18	S02	301.9739102	7	31	28.6	Male	Evening	4.609563352	3
19	S02	294.7954703	8	31	28.6	Male	Evening	5.252864702	2
20	S02	327.7613847	9	31	28.6	Male	Evening	7.260204074	2
21	S03	357.3727909	0	35	18.8	Male	Evening	7.582503285	2
22	S03	332.7920156	1	35	18.8	Male	Evening	3.215358165	1
23	S03	342.6608724	2	35	18.8	Male	Evening	7.567914291	2
24	S03	367.2285074	3	35	18.8	Male	Evening	5.060720166	2
25	S03	401.8518683	4	35	18.8	Male	Evening	6.760648028	4

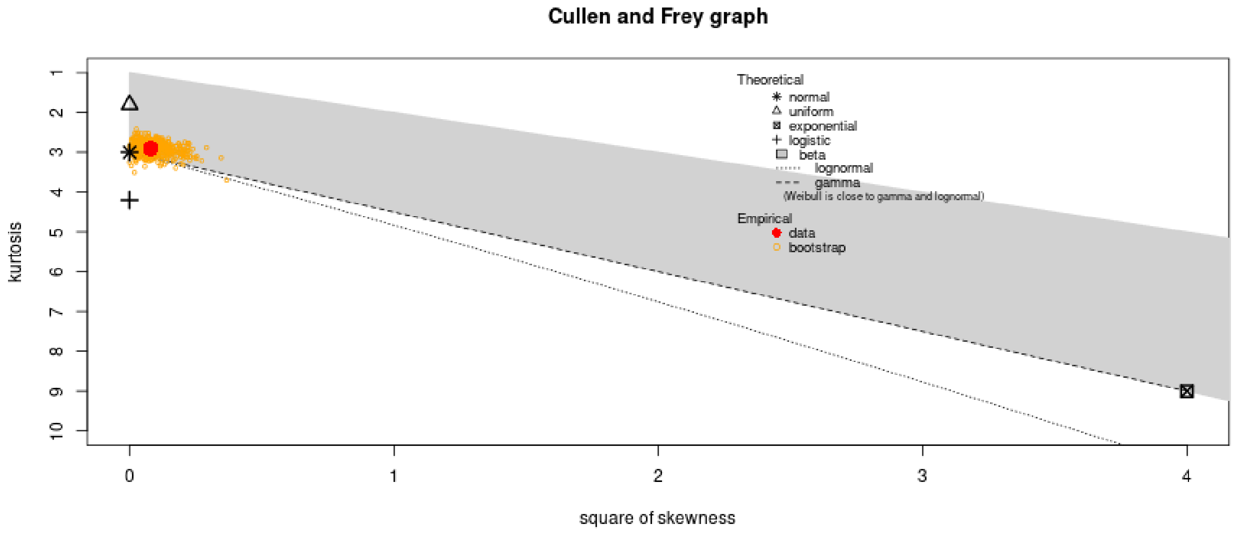
Fit Distribution Output

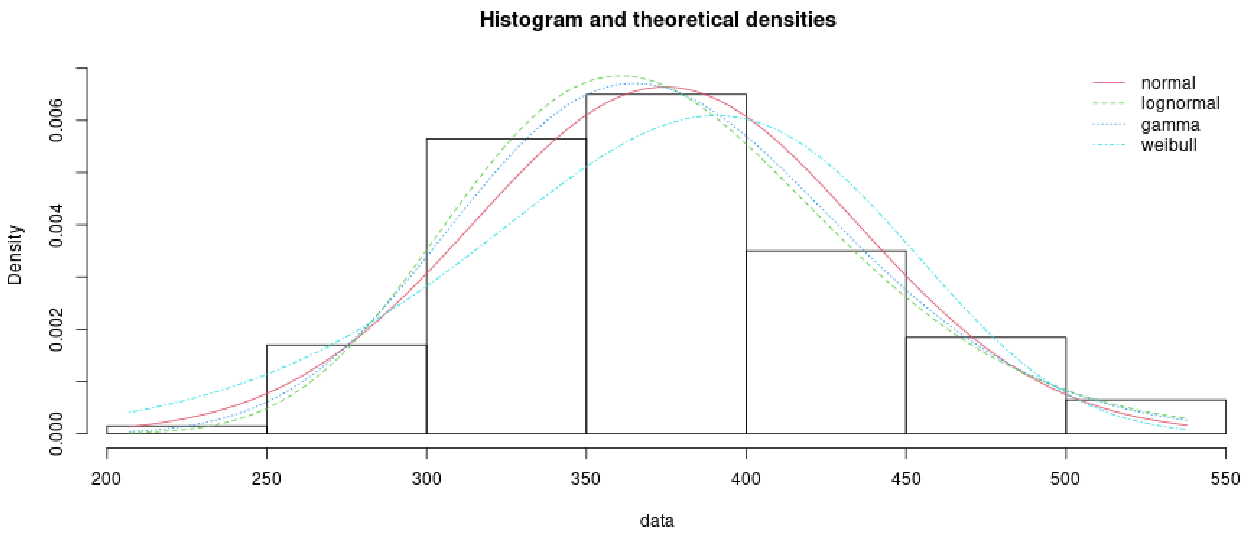
```
===== Dependent Variable: Reaction_time =====

---- Normal ----
Fitting of the distribution ' norm ' by matching moments
Parameters :
      estimate Std. Error
mean 374.62594   3.002912
sd    60.05824   2.123367
Loglikelihood:  -2205.701   AIC:  4415.403   BIC:  4423.386
Correlation matrix:
              mean            sd
mean  1.000000e+00 -7.499836e-07
sd   -7.499836e-07  1.000000e+00



---- Lognormal ----
Fitting of the distribution ' lnorm ' by matching moments
Parameters :
         estimate  Std. Error
meanlog 5.9132399 0.007966773
sdlog   0.1592993 0.005849832
Loglikelihood:  -2203.955   AIC:  4411.911   BIC:  4419.894
Correlation matrix:
             meanlog        sdlog
meanlog  1.000000000 -0.005753425
sdlog   -0.005753425  1.000000000



---- Gamma ----
Fitting of the distribution ' gamma ' by matching moments
Parameters :
       estimate  Std. Error
shape 38.909033 2.786435671
rate   0.103861 0.007484361
Loglikelihood:  -2202.777   AIC:  4409.555   BIC:  4417.538
Correlation matrix:
          shape      rate
shape 1.0000000 0.9937941
rate  0.9937941 1.0000000



---- Weibull ----
Fitting of the distribution ' weibull ' by maximum likelihood
Parameters :
        estimate Std. Error
shape   6.553063  0.2411781
scale 400.525215  3.2392306
Loglikelihood:  -2222.301   AIC:  4448.602   BIC:  4456.584
Correlation matrix:
          shape     scale
shape 1.0000000 0.3312924
scale 0.3312924 1.0000000



---- Goodness-of-fit ----
```

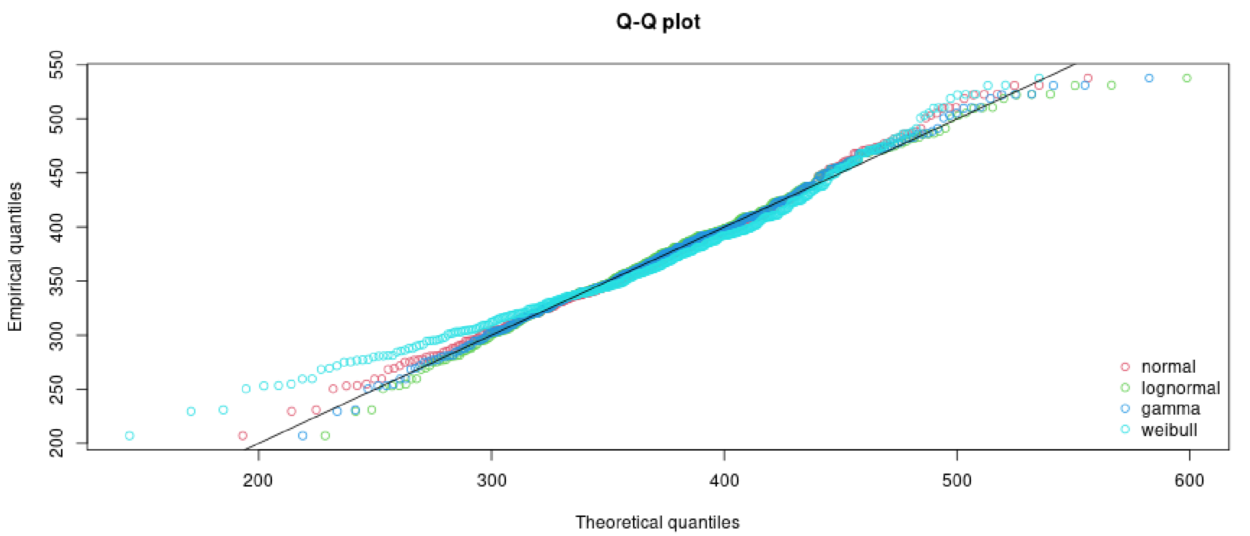
```
Goodness-of-fit statistics
                             1-mme-norm 2-mme-lnorm 3-mme-gamma 4-mle-weibull
Kolmogorov-Smirnov statistic 0.04199006  0.02235441  0.02641049     0.0737202
Cramer-von Mises statistic   0.16417978  0.03049209  0.03991641     0.6302238
Anderson-Darling statistic   1.04180442  0.26143932  0.28315921     4.0404176

Goodness-of-fit criteria
                               1-mme-norm 2-mme-lnorm 3-mme-gamma 4-mle-weibull
Akaike's Information Criterion   4415.403    4411.911    4409.555      4448.602
Bayesian Information Criterion   4423.386    4419.894    4417.538      4456.584
```

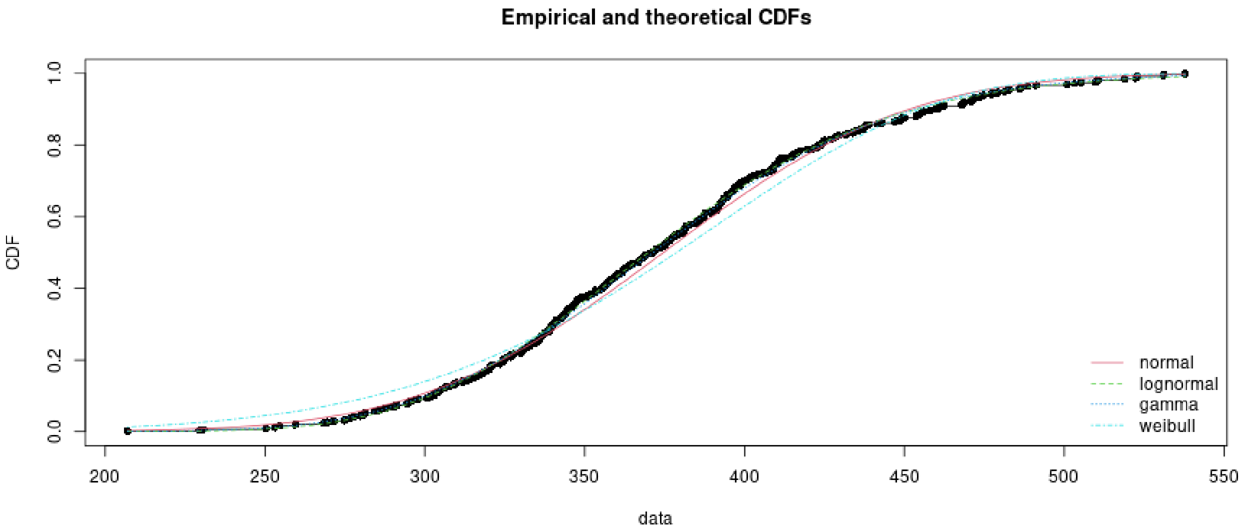
Fit Distribution Plots

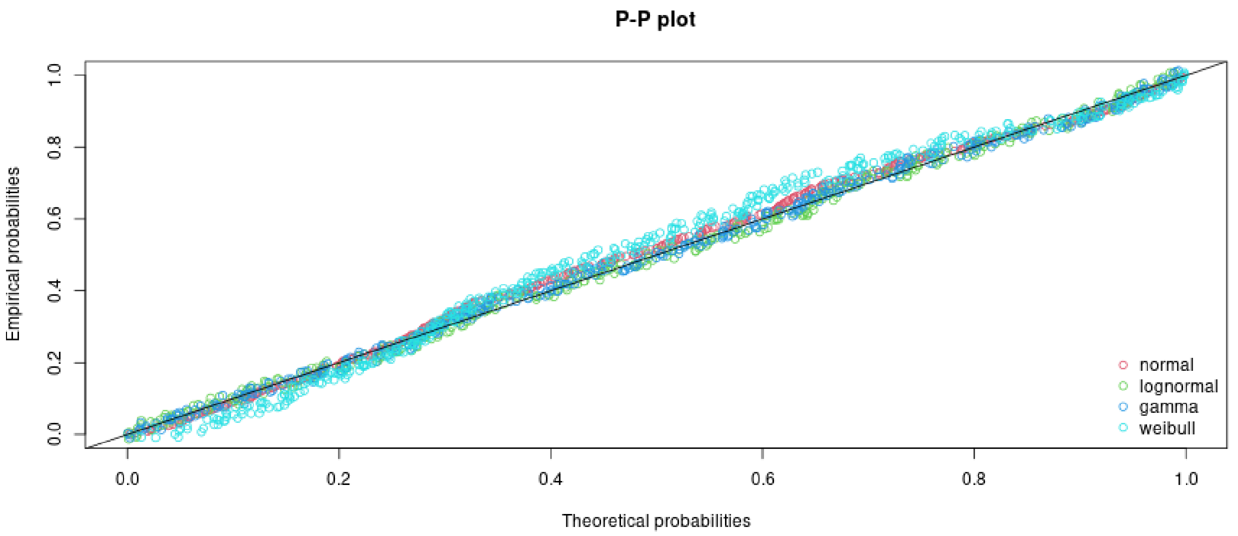
Distribution Fitting for Selected Dependent Variable












```
Goodness-of-fit statistics
                             1-mme-norm 2-mme-lnorm 3-mme-gamma 4-mle-weibull
Kolmogorov-Smirnov statistic 0.04199006  0.02235441  0.02641049     0.0737202
Cramer-von Mises statistic   0.16417978  0.03049209  0.03991641     0.6302238
Anderson-Darling statistic   1.04180442  0.26143932  0.28315921     4.0404176

Goodness-of-fit criteria
                               1-mme-norm 2-mme-lnorm 3-mme-gamma 4-mle-weibull
Akaike's Information Criterion   4415.403    4411.911    4409.555      4448.602
Bayesian Information Criterion   4423.386    4419.894    4417.538      4456.584
```

Model Output (afex)

```
--- Model: Days ---
Formula: Reaction_time ~ Days + age + bmi + Sex + Total_sleep_time + (1|Subject)
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
Formula: Reaction_time ~ Days + age + bmi + Sex + Total_sleep_time + (1 |      Subject)
   Data: data

REML criterion at convergence: 3776.5

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.62145 -0.64409 -0.01247  0.67402  2.91049

Random effects:
 Groups   Name        Variance Std.Dev.
 Subject  (Intercept) 1815.3   42.61
 Residual              547.9   23.41
Number of obs: 400, groups:  Subject, 40

Fixed effects:
                 Estimate Std. Error       df t value Pr(>|t|)
(Intercept)      252.1596    45.4285  38.4919   5.551 2.26e-06 ***
Days              11.3516     0.4388 359.4691  25.872  < 2e-16 ***
age                1.1522     0.4996  36.0089   2.306   0.0270 *
bmi                0.2699     1.3343  36.0131   0.202   0.8409
Sex1             -16.7713     7.2712  36.0331  -2.307   0.0269 *
Total_sleep_time   1.0140     1.0805 367.9936   0.938   0.3486
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) Days   age    bmi    Sex1
Days        -0.104
age         -0.562  0.004
bmi         -0.821  0.005  0.048
Sex1         0.149 -0.008 -0.236  0.019
Totl_slp_tm -0.178  0.371  0.011  0.014 -0.022

=== Effect Sizes ===
         Parameter Eta2_partial   CI      CI_low CI_high
1             Days  0.650563947 0.95 0.607171117       1
2              age  0.128806412 0.95 0.008528313       1
3              bmi  0.001135442 0.95 0.000000000       1
4              Sex  0.128740186 0.95 0.008527753       1
5 Total_sleep_time  0.002383416 0.95 0.000000000       1
```

ANOVA Output (afex)

```
--- ANOVA for: Days ---
Formula: Reaction_time ~ Days + age + bmi + Sex + Total_sleep_time + (1|Subject)
Mixed Model Anova Table (Type 3 tests, KR-method)

Model: Reaction_time ~ Days + age + bmi + Sex + Total_sleep_time + (1 |
Model:     Subject)
Data: df
                 num Df  den Df        F  Pr(>F)
Days                  1 359.441 669.1911 < 2e-16 ***
age                   1  35.980   5.3196 0.02696 *
bmi                   1  35.984   0.0409 0.84086
Sex                   1  36.004   5.3201 0.02695 *
Total_sleep_time      1 367.972   0.8791 0.34906
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Summary Table Output

Days

Model results: Days

num Df	den Df	F	Pr(>F)	Effect
1	359.4412	669.1911	0.0000	Days
1	35.9799	5.3196	0.0270	age
1	35.9841	0.0409	0.8409	bmi
1	36.0041	5.3201	0.0269	Sex
1	367.9721	0.8791	0.3491	Total_sleep_time

Performance

Instructions Data FitDist Output Fit Distribution Plots Model Output ANOVA Summary Table Performance Diagnostic flags

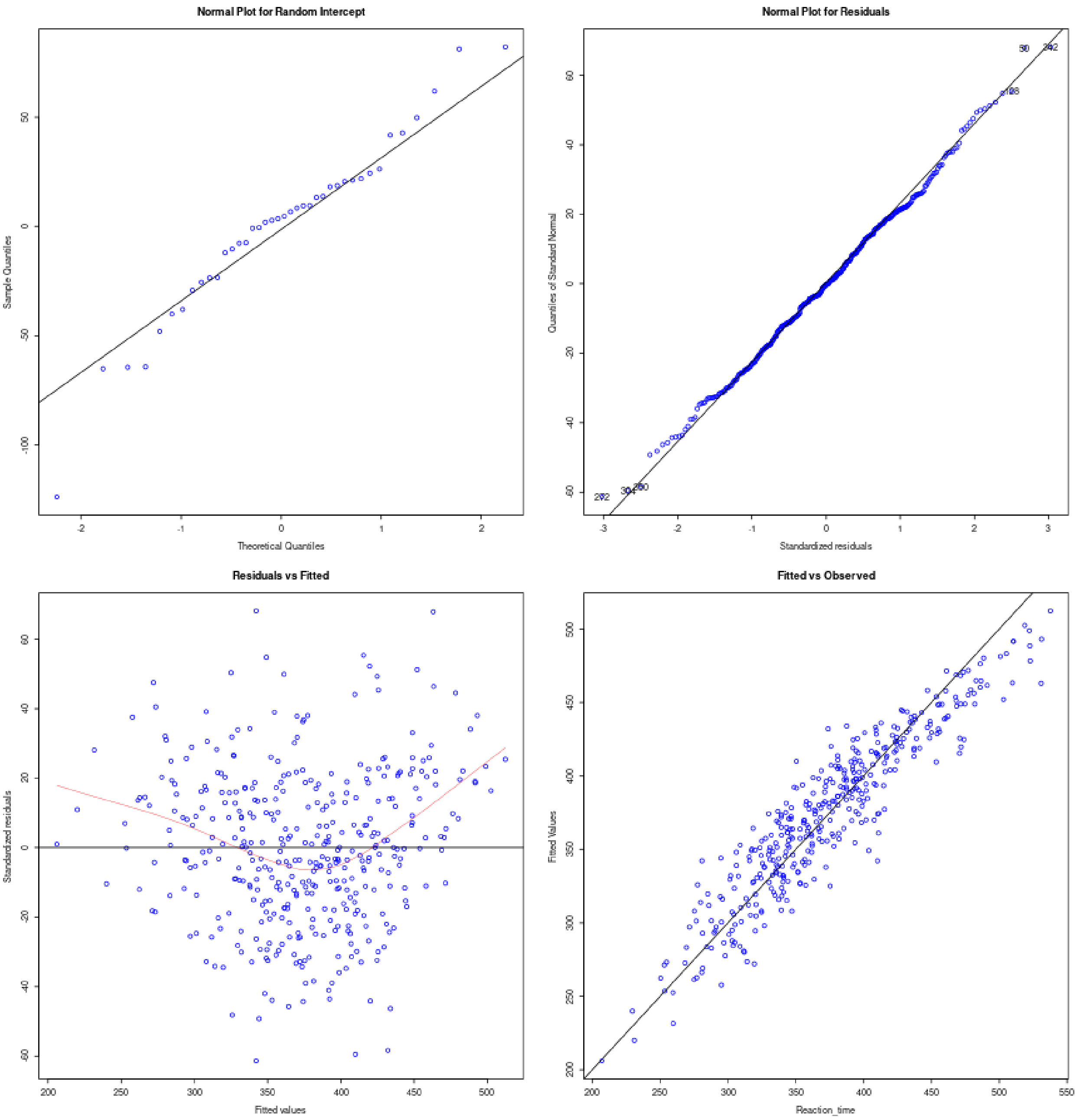
Post-hoc (EMMs) Summary Plots Diagnostics Auto Report Days

```
=== Diagnostic Flags ===
Singular fit: NO ✅
Convergence: OK
```

Performance metrics

```
# Indices of model performance

AIC    |   AICc |    BIC | R2 (cond.) | R2 (marg.) |   ICC |   RMSE |  Sigma
-----------------------------------------------------------------------------
3792.5 | 3792.9 | 3824.4 |      0.856 |      0.378 | 0.768 | 22.178 | 23.408
```

Model checks



Post hoc analysis

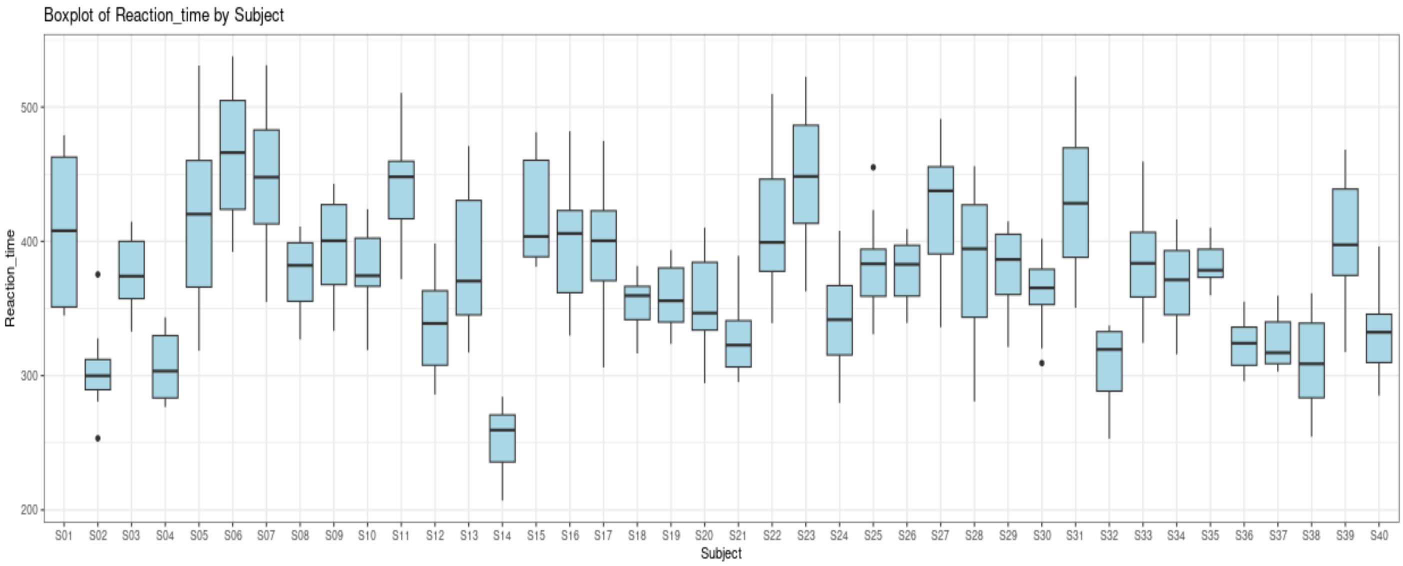
```
--- EMMs for: Days ---

Factor: Sex
 Sex    emmean    SE df lower.CL upper.CL
 Female    354 11.40 36      331      377
 Male      387  8.75 36      369      405

Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

Pairwise contrasts for: Sex
 contrast      estimate   SE df t.ratio p.value
 Female - Male    -33.5 14.5 36  -2.307  0.0269

Degrees-of-freedom method: kenward-roger
 contrast      estimate   SE df t.ratio p.value
 Female - Male    -33.5 14.5 36  -2.307  0.0269

Degrees-of-freedom method: kenward-roger
```

Summary Plots

Boxplots with Pairwise t-tests

Select Categorical IV:

Subject



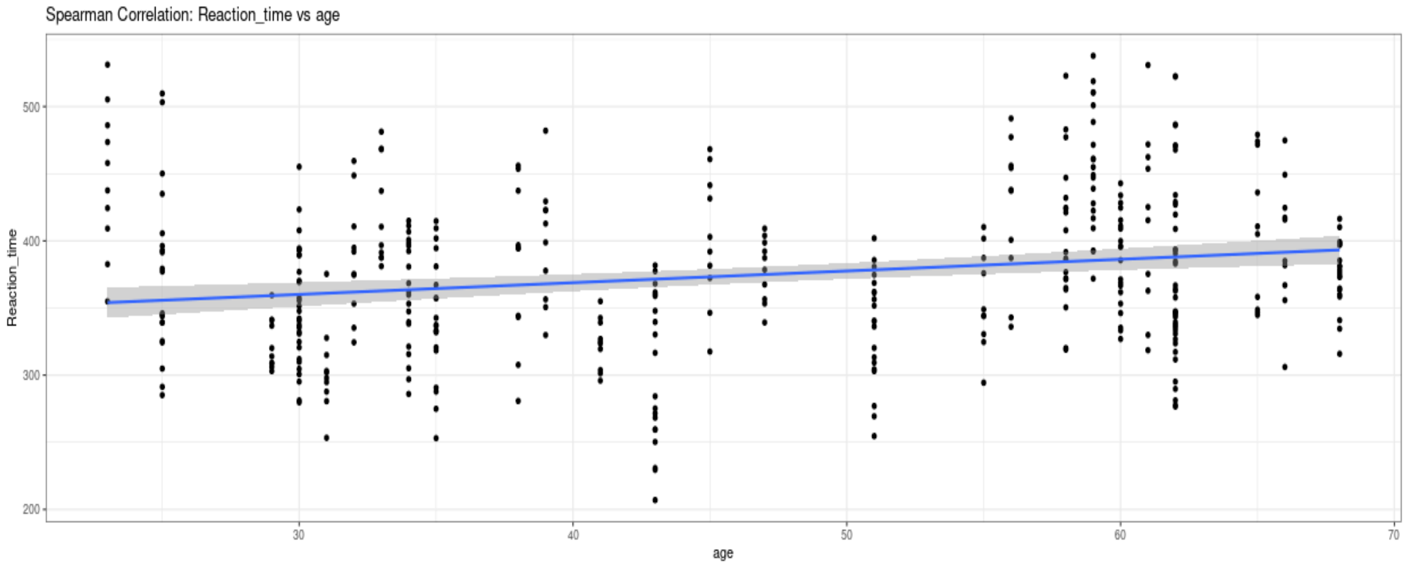
```
        Pairwise comparisons using t tests with pooled SD

data:  df[[input$y]] and df[[input$boxplot_cats]]

    S01     S02     S03     S04     S05     S06     S07     S08     S09     S10     S11     S12     S13     S14     S15     S16     S1
S02 3.6e-08 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S03 0.08656 0.00011 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S04 9.7e-08 0.84933 0.00023 -       -       -       -       -       -       -       -       -       -       -       -       -       -
S05 0.70090 4.3e-09 0.03617 1.2e-08 -       -       -       -       -       -       -       -       -       -       -       -       -
S06 0.00205 < 2e-16 2.1e-06 3.6e-16 0.00683 -       -       -       -       -       -       -       -       -       -       -       -
S07 0.03605 1.0e-13 0.00016 3.7e-13 0.08632 0.31754 -       -       -       -       -       -       -       -       -       -       -
S08 0.09665 8.7e-05 0.95786 0.00018 0.04108 2.7e-06 0.00019 -       -       -       -       -       -       -       -       -       -
S09 0.45393 1.6e-06 0.33329 3.8e-06 0.25750 0.00014 0.00457 0.36031 -       -       -       -       -       -       -       -       -
S10 0.09133 9.7e-05 0.97962 0.00021 0.03848 2.4e-06 0.00017 0.97823 0.34618 -       -       -       -       -       -       -       -
S11 0.05047 2.7e-13 0.00027 9.3e-13 0.11542 0.25395 0.88738 0.00033 0.00700 0.00029 -       -       -       -       -       -       -
S12 0.00024 0.05507 0.04724 0.08368 5.2e-05 4.0e-11 1.3e-08 0.04169 0.00328 0.04448 2.9e-08 -       -       -       -       -       -
S13 0.29172 6.5e-06 0.50801 1.5e-05 0.15064 4.0e-05 0.00171 0.54242 0.75962 0.52450 0.00272 0.00832 -       -       -       -       -
S14 1.4e-15 0.00678 1.2e-10 0.00380 < 2e-16 < 2e-16 < 2e-16 8.5e-11 2.5e-13 1.0e-10 < 2e-16 4.7e-06 1.8e-12 -       -       -       -
S15 0.46465 5.9e-10 0.01474 1.8e-09 0.72837 0.01816 0.17084 0.01701 0.13930 0.01580 0.21930 1.2e-05 0.07463 < 2e-16 -       -       -
S16 0.61925 4.6e-07 0.22282 1.2e-06 0.37849 0.00036 0.00966 0.24346 0.80092 0.23264 0.01437 0.00144 0.57681 4.5e-14 0.21975 -       -
S17 0.59629 5.4e-07 0.23550 1.4e-06 0.36100 0.00032 0.00879 0.25695 0.82640 0.24570 0.01313 0.00160 0.59942 5.6e-14 0.20770 0.97380 -
S18 0.00405 0.00642 0.24116 0.01114 0.00115 4.9e-09 9.1e-07 0.22066 0.03280 0.23109 1.8e-06 0.41449 0.06709 8.7e-08 0.00033 0.01712 0
S19 0.00904 0.00280 0.36531 0.00508 0.00280 2.1e-08 3.2e-06 0.33804 0.06157 0.35196 6.2e-06 0.27885 0.11751 2.1e-08 0.00087 0.03405 0
S20 0.00561 0.00466 0.28623 0.00822 0.00165 8.8e-09 1.5e-06 0.26309 0.04240 0.27488 3.0e-06 0.35663 0.08438 5.0e-08 0.00049 0.02265 0
S21 2.0e-05 0.19200 0.00947 0.26473 3.5e-06 7.9e-13 4.0e-10 0.00813 0.00039 0.00880 9.2e-10 0.53731 0.00117 6.8e-05 6.7e-07 0.00015 0
S22 0.74100 5.9e-09 0.04116 1.7e-08 0.95726 0.00582 0.07700 0.04664 0.28065 0.04374 0.10361 6.5e-05 0.16639 < 2e-16 0.68851 0.40815 0
S23 0.02058 2.3e-14 6.4e-05 8.3e-14 0.05299 0.43628 0.82478 8.0e-05 0.00226 7.1e-05 0.71661 3.9e-09 0.00080 < 2e-16 0.11185 0.00502 0
S24 0.00038 0.04133 0.06257 0.06406 8.6e-05 8.5e-11 2.6e-08 0.05554 0.00481 0.05908 5.6e-08 0.90219 0.01181 2.7e-06 2.0e-05 0.00216 0
S25 0.18317 2.2e-05 0.70054 4.9e-05 0.08665 1.2e-05 0.00065 0.74006 0.55964 0.71955 0.00108 0.01803 0.78140 1.1e-11 0.03958 0.40356 0
S26 0.11750 5.9e-05 0.88131 0.00013 0.05153 4.2e-06 0.00028 0.92315 0.41310 0.90150 0.00047 0.03299 0.60813 4.7e-11 0.02196 0.28455 0
S27 0.42966 4.2e-10 0.01254 1.3e-09 0.68480 0.02120 0.18983 0.01451 0.12435 0.01346 0.24203 9.2e-06 0.06563 < 2e-16 0.95324 0.19857 0
S28 0.14859 3.6e-05 0.78662 7.8e-05 0.06777 7.3e-06 0.00043 0.82753 0.48568 0.80633 0.00072 0.02430 0.69551 2.2e-11 0.02993 0.34261 0
S29 0.14064 4.0e-05 0.80903 8.8e-05 0.06355 6.4e-06 0.00039 0.85021 0.46772 0.82888 0.00065 0.02617 0.67419 2.7e-11 0.02783 0.32809 0
S30 0.01228 0.00197 0.42528 0.00364 0.00395 3.8e-08 5.3e-06 0.39530 0.07807 0.41063 1.0e-05 0.23373 0.14494 1.2e-08 0.00127 0.04419 0
S31 0.26811 6.2e-11 0.00495 2.0e-10 0.46910 0.04668 0.32035 0.00581 0.06387 0.00535 0.39401 2.1e-06 0.03104 < 2e-16 0.70629 0.10905 0
S32 1.3e-07 0.79925 0.00029 0.94869 1.8e-08 5.7e-16 5.6e-13 0.00024 5.1e-06 0.00026 1.4e-12 0.09577 2.0e-05 0.00310 2.6e-09 1.6e-06 1
S33 0.26289 8.7e-06 0.55076 2.0e-05 0.13301 3.0e-05 0.00137 0.58657 0.71038 0.56794 0.00220 0.01004 0.94787 2.8e-12 0.06465 0.53303 0
S34 0.03543 0.00048 0.69464 0.00095 0.01301 3.1e-07 3.2e-05 0.65604 0.17414 0.67587 5.7e-05 0.11090 0.29191 1.2e-09 0.00472 0.10739 0
S35 0.16423 2.8e-05 0.74561 6.3e-05 0.07621 9.2e-06 0.00053 0.78592 0.51990 0.76502 0.00087 0.02114 0.73562 1.6e-11 0.03420 0.37059 0
S36 6.6e-06 0.29019 0.00453 0.38534 1.1e-06 1.5e-13 9.1e-11 0.00385 0.00015 0.00419 2.1e-10 0.38742 0.00049 0.00018 2.0e-07 5.6e-05 6
S37 7.8e-06 0.27293 0.00511 0.36453 1.3e-06 2.0e-13 1.2e-10 0.00434 0.00018 0.00472 2.7e-10 0.40901 0.00056 0.00016 2.4e-07 6.6e-05 7
S38 2.1e-07 0.73577 0.00039 0.88273 2.8e-08 1.0e-15 9.7e-13 0.00032 7.5e-06 0.00036 2.4e-12 0.11341 2.8e-05 0.00237 4.1e-09 2.3e-06 2
S39 0.74348 2.0e-07 0.16508 5.1e-07 0.47697 0.00067 0.01551 0.18170 0.67314 0.17296 0.02260 0.00080 0.46684 1.4e-14 0.29009 0.86523 0
S40 9.7e-05 0.09154 0.02678 0.13404 2.0e-05 9.3e-12 3.7e-09 0.02339 0.00153 0.02510 8.2e-09 0.81608 0.00413 1.3e-05 4.2e-06 0.00064 0
    S25     S26     S27     S28     S29     S30     S31     S32     S33     S34     S35     S36     S37     S38     S39
S02 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S03 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S04 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S05 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S06 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S07 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S08 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S09 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S10 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S11 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S12 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S13 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S14 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S15 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S16 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S17 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S18 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S19 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S20 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S21 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S22 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S23 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S24 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S25 -       -       -       -       -       -       -       -       -       -       -       -       -       -       -
S26 0.81396 -       -       -       -       -       -       -       -       -       -       -       -       -       -
S27 0.03433 0.01882 -       -       -       -       -       -       -       -       -       -       -       -       -
S28 0.90930 0.90337 0.02581 -       -       -       -       -       -       -       -       -       -       -       -
S29 0.88634 0.92639 0.02396 0.97685 -       -       -       -       -       -       -       -       -       -       -
S30 0.23755 0.34397 0.00104 0.28574 0.29901 -       -       -       -       -       -       -       -       -       -
S31 0.01506 0.00774 0.75032 0.01098 0.01011 0.00033 -       -       -       -       -       -       -       -       -
S32 6.4e-05 0.00016 1.8e-09 0.00010 0.00011 0.00444 2.9e-10 -       -       -       -       -       -       -       -
S33 0.83201 0.65459 0.05667 0.74440 0.72256 0.16376 0.02633 2.6e-05 -       -       -       -       -       -       -
S34 0.43721 0.58795 0.00394 0.50725 0.52600 0.68549 0.00140 0.00119 0.32281 -       -       -       -       -       -
```

```
S35 0.95202 0.86096 0.02957 0.95714 0.93404 0.26224 0.01276 8.1e-05 0.78540 0.47349 -       -       -       -       -
S36 0.00130 0.00284 1.4e-07 0.00191 0.00210 0.04030 2.7e-08 0.42149 0.00062 0.01423 0.00160 -       -       -       -
S37 0.00148 0.00321 1.8e-07 0.00217 0.00238 0.04419 3.3e-08 0.39950 0.00071 0.01581 0.00182 0.96911 -       -       -
S38 8.9e-05 0.00022 2.9e-09 0.00014 0.00016 0.00573 4.7e-10 0.93373 3.7e-05 0.00157 0.00011 0.47104 0.44758 -       -
S39 0.31504 0.21519 0.26424 0.26341 0.25126 0.02922 0.15169 7.0e-07 0.42782 0.07528 0.28699 2.8e-05 3.2e-05 1.0e-06 -
S40 0.00947 0.01816 3.2e-06 0.01306 0.01414 0.15486 7.0e-07 0.15149 0.00505 0.06795 0.01124 0.52741 0.55300 0.17654 0.00034

P value adjustment method: none
```

Spearman Correlation Plots

Select Numeric IV:

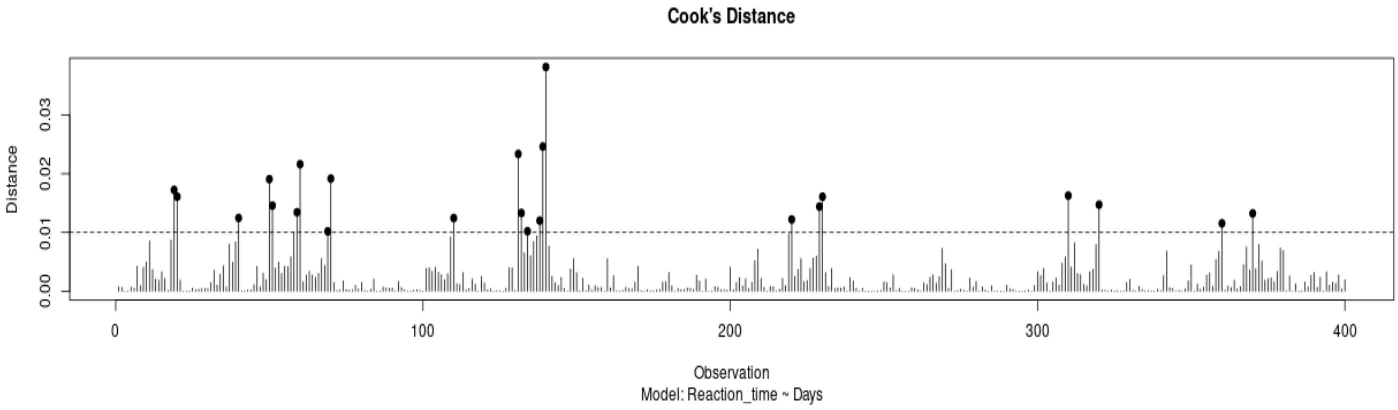
age



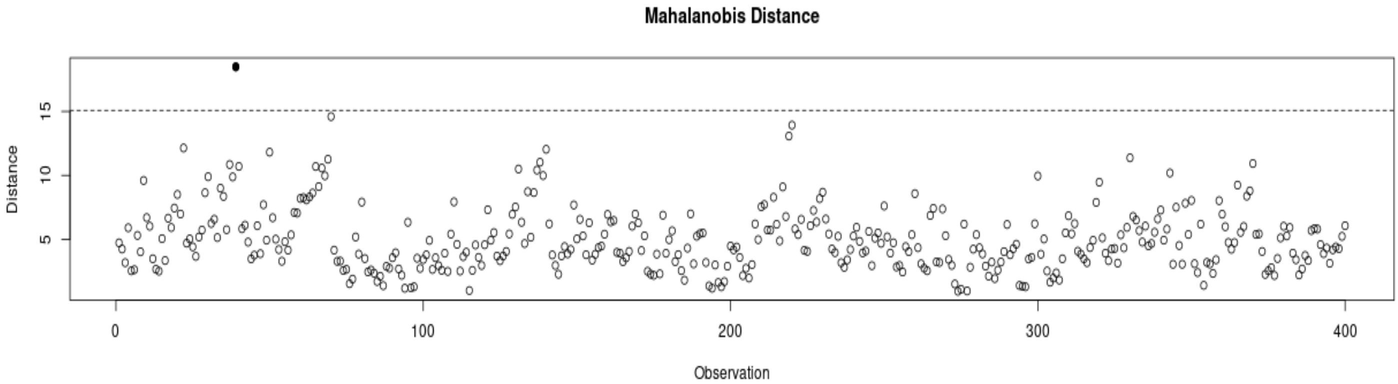

```
$Spearman

        Spearman's rank correlation rho

data:  df[[input$corr_iv]] and df[[input$y]]
S = 8548789, p-value = 6.378e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
     rho
0.198546


$Regression
                       Estimate Std. Error   t value      Pr(>|t|)
(Intercept)         333.9753942 10.1767494 32.817492 2.867140e-115
df[[input$corr_iv]]   0.8718617  0.2089182  4.173221  3.690703e-05
```

Outlier Diagnostic Output

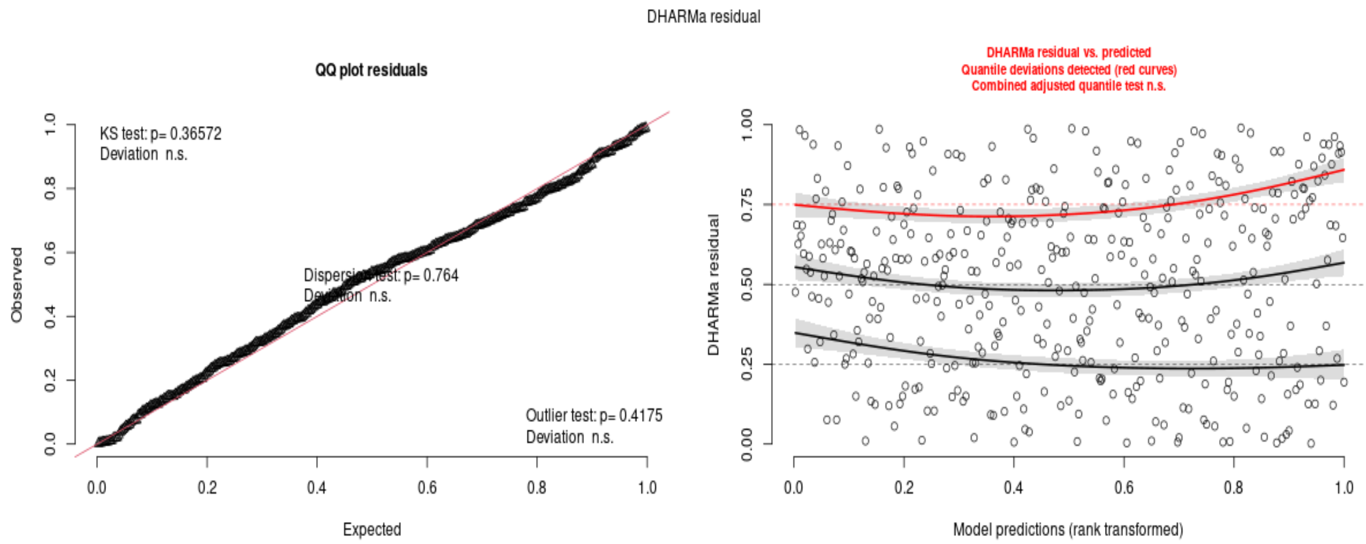
Cook's Distance [Default cutoff]



Mahalanobis Distance [Default cutoff]



DHARMa Residual Diagnostics



```
=== DHARMa Diagnostics ===

        Asymptotic one-sample Kolmogorov-Smirnov test

data:  simulationOutput$scaledResiduals
D = 0.046, p-value = 0.3657
alternative hypothesis: two-sided


        DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated

data:  simulationOutput
dispersion = 0.93505, p-value = 0.764
alternative hypothesis: two.sided


        DHARMa outlier test based on exact binomial test with approximate expectations

data:  sim_res
outliers at both margin(s) = 0, observations = 400, p-value = 0.4175
alternative hypothesis: true probability of success is not equal to 0.003992016
95 percent confidence interval:
 0.000000000 0.009179805
sample estimates:
frequency of outliers (expected: 0.00399201596806387 )
                                                     0


Number of outliers: 0
```